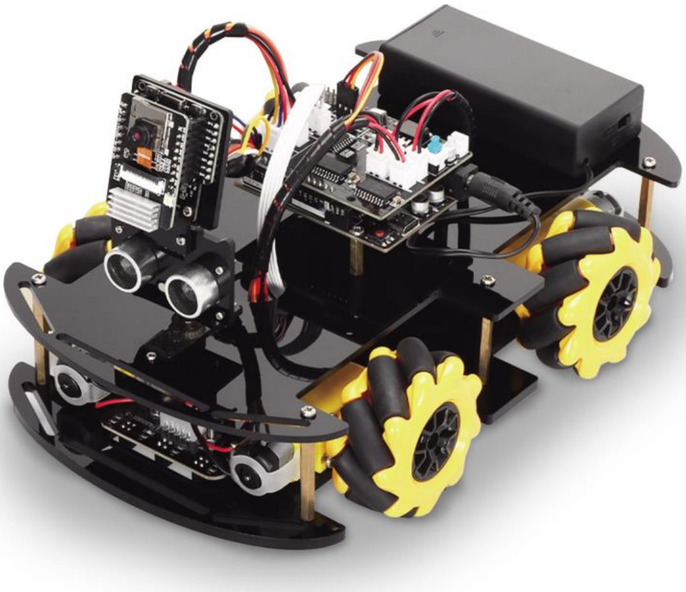
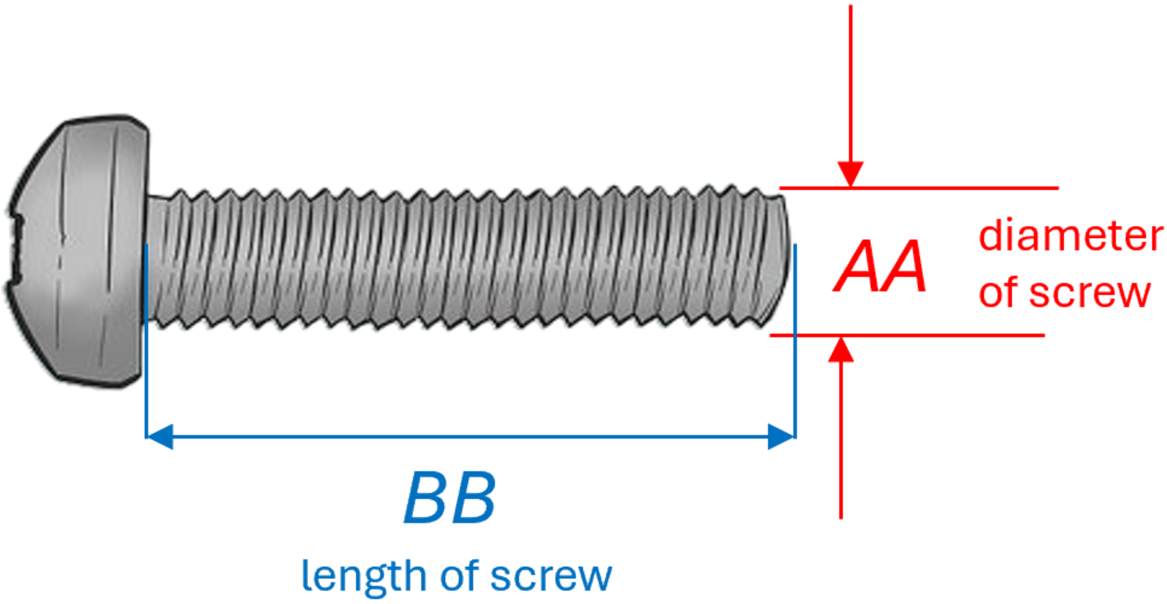
SMART ROBOT
PROGRAMMING LEARNING

Manual & Tutorial





MAA x BB

M stands for metric, all dimensions in millimetres

e.g. M3 x 20 : 3mm diameter, 20mm length

Types of Screws used:

Product Parts

M1.6 x 10 Round Head Screws (4)

M1.6 Nuts (4)

M2 x 10 Round Head Screws (6)

M2 Nuts (6)

M2.5 x 20 Round Head Screws (4)

M3 x 8 Round Head Screws (28)

M3 x 10 Flat Head Screws (4)

M3 x 16 Round Head Screws (2)

M3 x 30 Round Head Screws (8)

Product Parts

M3 Nuts (16)

M3 x 15 Copper Pillar (4)

M3 x 40 Copper Pillar (6)

Motor Bracket (4)

Phillips Screwdriver

Slotted Screwdriver

4P-1P Connection Cable 20cm

4P Connection Cable 20cm

5P Connection Cable 20cm

Product Parts

Development Board

Expansion Board

ESP32-CAM Board

ESP32 Expansion Board

Ultrasound Module

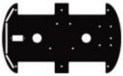
Tracking Module

Servo

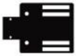
Motor (4)

Battery Box

Product Parts



Acrylic Top Plate



ESP32 Expansion Board Bracket

Ultrasonic Module Bracket

Acrylic Base Plate

Wheel & Coupling (4)

Type-C USB Cable

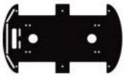
Winding Tube (2)

Installation of the Tracking Module

① M3 x 16 Round Head Screws (4)

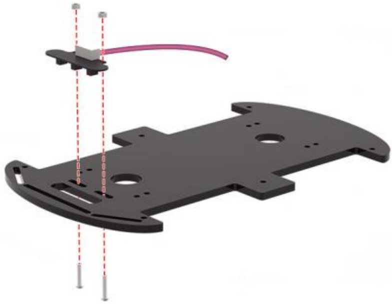
② M3 Nuts (4)

③ 5P Connection Cable 20cm



④ Acrylic Base Plate

⑤ Tracking Module



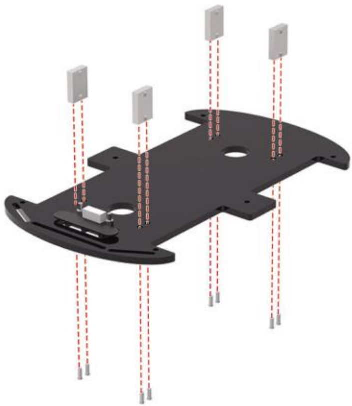
Secure the tracking module to the acrylic base plate as shown in the picture and plug the 5P connection cable into the tracking module.

Note: Two additional nuts are required between the module and the base plate.

Installation of the Motor Brackets

① M3 x 8 Round Head Screws (8)

② Motor Bracket (4)



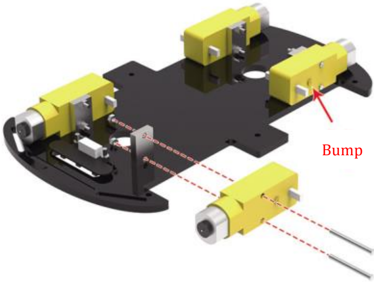
Attach the motor bracket to the acrylic base plate as shown in the picture.

Installing the Motor

① M3 x 30 Round Head Screws (8)

② M3 Nuts (8)

③ Motor (4)

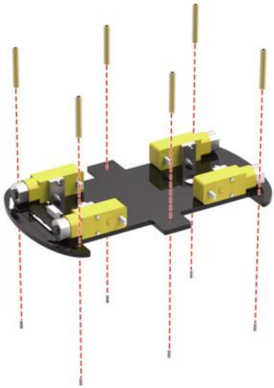


Secure the 4 motors to the motor bracket as shown in the picture.
Note: The side of the 4 motors with the bumps faces outward.

Installation of the Copper Pillar

① M3 x 8 Round Head Screws (6)

② M3 x 40 Copper Pillar (6)

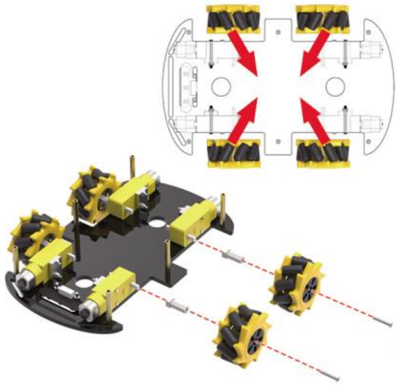


Attach the copper pillar to the acrylic base plate as shown in the picture.

Installation of Wheels

① Wheel & Coupling (4)

② M2.5 x 20 Round Head Screws (4)



Secure the coupling to the motor as shown in the picture, then secure the wheel to the coupling.

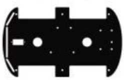
Note: Take note of the wheel placement.

Installation the Servo

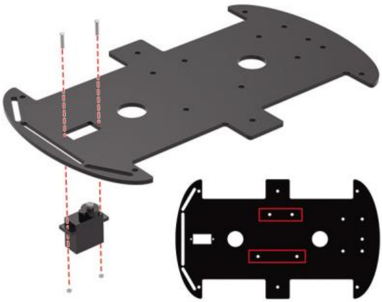
① M2 x 10 Round Head Screws (2)

② M2 Nuts (2)

③ Servo



④ Acrylic Top Plate



This side faces up

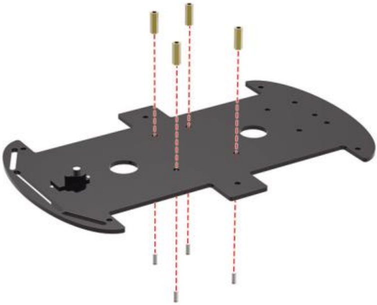
Attach the servo to the underside of the acrylic top plate as shown in the pictures.

Note: Take note of the front and back of the acrylic top plate.

Install the Copper Pillar

① M3 x 8 Round Head Screws (4)

② M3 x 15 Copper Pillar (4)



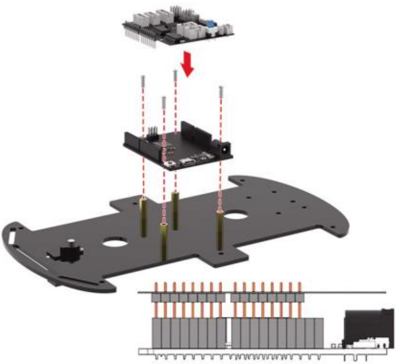
Secure the copper pillar to the acrylic top plate as shown in the picture.

Install the Development Board & Expansion Board

① M3 x 8 Round Head Screws (4)

② Development Board

③ Expansion Board



Secure the development board to the copper pillar as shown in the picture and then plug the expansion board into the development board.

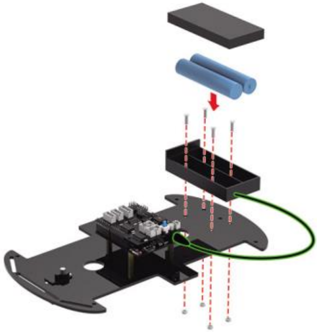
Note: Don't plug the board's pins in the wrong way.

Installing the Battery Box

① M3 x 10 Flat Head Screws (4)

② M3 Nuts (4)

③ Battery Box



First, fix the battery box on the acrylic top plate, then align the positive and negative terminals of batteries and put them into the battery box, finally plug the power cord into the development board.

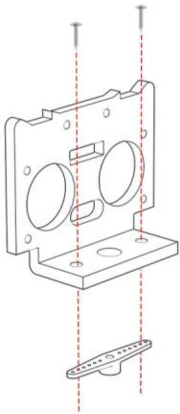
Note: The product does not include batteries, you are advised to prepare two 3.7V batteries.

Assembling the Ultrasonic Module Bracket and Servo Arm

① Servo Tip Screws (2)

② Servo Arm (1)

③ Ultrasonic Module Bracket

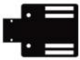


Attach the ultrasonic module bracket to the servo arm as shown in the picture.

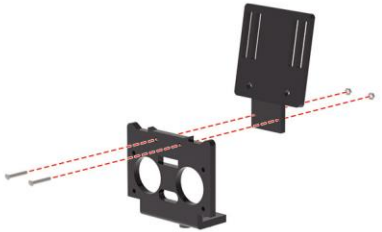
Installation of ESP32 Expansion Board Bracket

① M2 x 10 Round Head Screws (2)

② M2 Nuts (2)



③ ESP32 Expansion Board Bracket



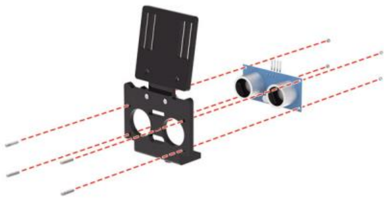
Attach the ESP32 expansion board bracket to the ultrasonic module bracket as shown in the picture.

Installation of Ultrasonic Module

① M1.6 x 10 Round Head Screws (4)

② M1.6 Nuts (4)

③ Ultrasonic Module



Secure the ultrasonic module to the ultrasonic module bracket.

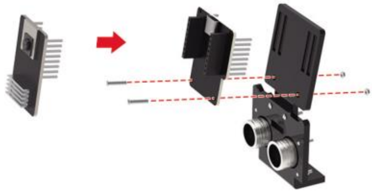
Assembling the ESP32-CAM and ESP32 Expansion Boards

① M2 x 10 Round Head Screws (2)

② M2 Nuts (2)

③ ESP32-CAM Board

④ ESP32 Expansion Board



Secure the ESP32 expansion board to the bracket as shown in the picture, then plug the ESP32-CAM into the ESP32 expansion board.

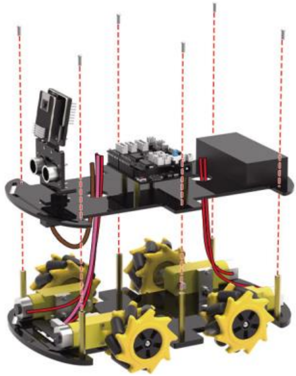
Installation Head

① Round Head Servo Screws (1)

Secure the head of the car to the servo as shown in the picture.

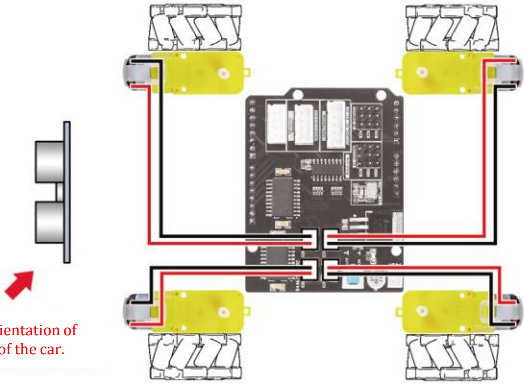
Assemble the Whole Car

① M3 x 8 Round Head Screws (6)



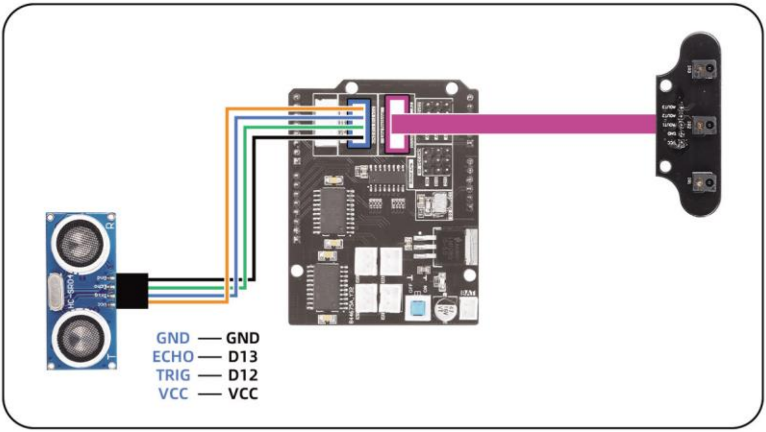
As shown in the picture, first pass the bottom wire through the round hole in the acrylic top plate, then secure the acrylic top plate to the copper pillar.

Wiring

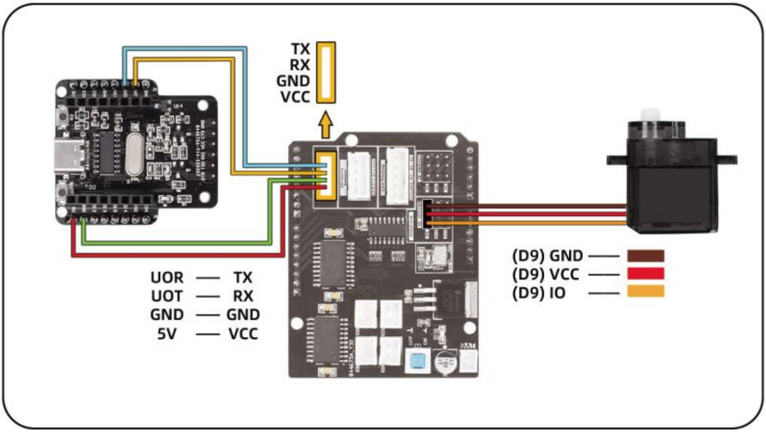


Note the orientation of the front of the car.

Wiring



Wiring



Assembly Completed

产品基本参数

Working voltage: 5V

Input voltage: 7-9V

Maximum output current: 2A

Maximum power consumption: 25W (T=75°C)

Motor speed: 5V 200 rpm

Ultrasonic sensing angle: < 15 °

Ultrasonic detection distance: 2cm~400cm

Control range of cell phone: 10~50 meters

Servo angle range: 0~180°